

Research Article

Smart Healthcare: Disease Prediction Using the Cuckoo-Enabled Deep Classifier in IoT Framework

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The Internet of Things (IoT) is commonly employed to detect different kinds of diseases in the health sector. Presently, disease detection is performed using MRI images, X-rays, CT scans, and so on for diagnosing the diseases. The manual detection process is found to be time-consuming and may result in detection errors that affect the diagnosis. Hence, there is a need for an automatic system for which the deep learning methods gain a major interest. Hence, the idea to combine deep learning and disease prediction to effectively predict the disease is initiated. In this research, the deep learning method is combined with deep learning for the effective prediction of diseases, where the IoT network is employed in the data collection from the patients. The proposed cuckoo-based deep convolutional long-short term memory (deep convLSTM) classifier is employed for disease prediction, where the cuckoo search optimization is utilized for tuning the deep convLSTM classifier. The proposed method is compared with the conventional methods, and it achieved a training percentage of 97.591%, 95.874%, and 97.094%, respectively, for accuracy, sensitivity, and specificity. The comparative analysis proved that the proposed method obtained higher accuracy than other methods.

1. Introduction

IoT [9] is an emerging new technology for the upcoming later generation technology that interconnects specific smart objects with the system. IoT is a combination of various objects that are fixed invisibly around the globe [5, 10]. Health monitoring (HM) is one of the major research sectors in terms of skin attachable electronic devices. Smart HM means controlling and computing of the remote HM devices along with IoT [5, 11]. Employing advanced modern hardware sensors in the medical sector helps to generate a new technology called the Internet of medical things (IoMT) [4]. The aim of smart healthcare is the efficient monitoring of the patients through efficient patient information sharing,

emergency servicing, patient monitoring, and so on to reduce the risk of the patient's life. The smart healthcare field imposes information technology for developing advanced applications to improve medication and diagnostic procedures [25, 26]. Kumar et al. gave many solutions for detecting the object from images using machine learning algorithms [27–30]. Modern methods and research theories are the majority structures that produce large amounts of digital information [4]. Diabetes is a long-lasting disease all over the world, which begins when the human body loses the capability to synthesize a hormone called insulin. The world health organization (WHO) announced that diabetes was a chronic disease that resulted in 1.6 million deaths in the year 2016 [6, 7]. Patients affected with diabetes produce

high levels of glucose in the blood that damages the organs of the body [6]. The features of the healthcare models comprise insufficient medical data, clogged information, warnings in data, and so on. Hence, to overcome this challenge, skin attachable sensors that are associated with the Internet of Things (IoT) with big data emerged in the present world [3, 8]. When compared to the olden days, nowadays IoT and AI methods help to predict the various health issues accurately [6].

In the smart healthcare scenario, the patient's information is stored in the server, and as per the requirement, it can be retrieved and the necessary diagnosis can be performed. For the diagnosis of the disease, various deep learning techniques are implemented on the basis of the structural data of the anatomical system known as artificial neural networks (ANNs). Widely known deep learning methods, such as recurrent neural networks (RNN), deep belief networks (DBN), convolutional neural networks (CNN), hybrid neural networks, and deep neural networks (DNN), are employed to process the health data. In the last few years, deep learning methods have gained attention in the field of IoT to enhance security by finding solutions to threats [2]. The accuracy of the deep learning techniques can be further enhanced by using the optimization algorithms such as cuckoo optimization [17, 20], particle swarm optimization [18], crow search algorithm [24], and so on.

1.1. Internet of Health Things. Recently, IoT (Internet of Things) has been said to be one of the promising technologies from which healthcare received significant benefits, as illustrated in Figure 1. In such a way, [31] presented a comprehensive review on IoT in healthcare, which could be determined as IoHT (Internet of Health Things). Furthermore, this study identifies technical advances and its corresponding limitations to be overcome. The study also suggested that further studies are more essential in this field for improvising the current challenges faced by IoHT. Finally, the study stated that this comprehensive review will be a great information source to the healthcare providers, technology specialists, researchers, as well as general population for improvising IoHT. An efficient cryptosystem for securing the transmission of MRI images in the IoHT (Internet of Health Things) environment has been reported by [32]. This study investigated the dynamic of 2-dimension trigonometric map, which has infinite solutions. Furthermore, the study utilized phase portrait, bifurcation diagram, and Lyapunov exponent for demonstrating the complex dynamics of the map. From the performance analysis of the suggested cryptosystem, it can be concluded that it is highly secure and can be utilized in the internet of health things for the transmission of medical images in a secured manner.

Normally, deep learning techniques are implemented to improve the performance to predict the security threats in IoT [2]. The PIMA dataset is used for disease prediction, which is subjected to pre-processing for eliminating the unwanted noises in the data. The preprocessed data are then fed to the feature extractor for extracting the required features, which are then passed to the classifier that classifies

the data. The biases and weights of the classifier are tuned optimally using the cuckoo search optimization, which altogether efficiently predicts the disease.

1.2. Proposed Cuckoo Search-Based Conv LSTM Classifier.

The proposed disease prediction model is designed based on the cuckoo search-based deep LSTM classifier such that the internal modal parameters of the classifier are tuned using the cuckoo search optimization, which is a nature-inspired algorithm that chooses the best parameter set for predicting the diseases.

The accomplishment of the research is as follows: section 1 enumerates the purpose for the disease prediction, section 2 explains the existing works and the vulnerabilities, section 3 explains the methodology for the proposed method, section 4 illustrates the results and the comparative analysis of the proposed over the existing conventional methods, and section 5 deliberates the conclusion of the research work.

2. Motivation

In this segment, the reviews of the existing methods and the vulnerabilities of the existing methods are explained in detail.

2.1. Literature Review. The review of various pieces of research is as follows: Huma Naz and Sachin Ahuja [1] implemented a machine learning algorithm using the PIMA dataset. This method used different classifiers to predict the disease accurately but failed to diagnose the disease in the early stage. Usman Ahmad and others. [2] employed an artificial neural network (ANN) that effectively predicted the disease in the case of small datasets. Yet, this method is ineffective for larger datasets. Nithya Rekha Sivakumar and Faten Khalid Diaaldin Karim [3] explained the equidistant heuristic and duplex deep neural network (EH-DDNN) that enhanced the detection accuracy of diabetics. Though this method outperformed the existing methods, the prediction time must be lowered. Romany Fouad Mansour and others. [4] illustrated a new AI and IoT convergence-based disease diagnosis model for a smart healthcare system. This method had a higher prediction accuracy regarding the diagnoses of heart diseases and diabetes, however, there is no feature selection method, and hence, it had complex computations. Simanta Shekhar Sarmah [5] explained that the deep learning modified neural network (DLMNN) achieved higher data security, however, the disease detection rate had to be enhanced. The prediction based on machine learning using optimization was developed by [17]. The developed method obtained better accuracy in prediction with minimal computational cost, however, the slow convergence is considered to be the drawback of the method. The fuzzy-based classification using optimization was developed by [18] for the pattern classification. The minimal computation time with enhanced classification accuracy was obtained but failed to consider the preprocessing or postprocessing technique that enhances the quality of output. Disease prediction using the fuzzy technique was modeled by [19],

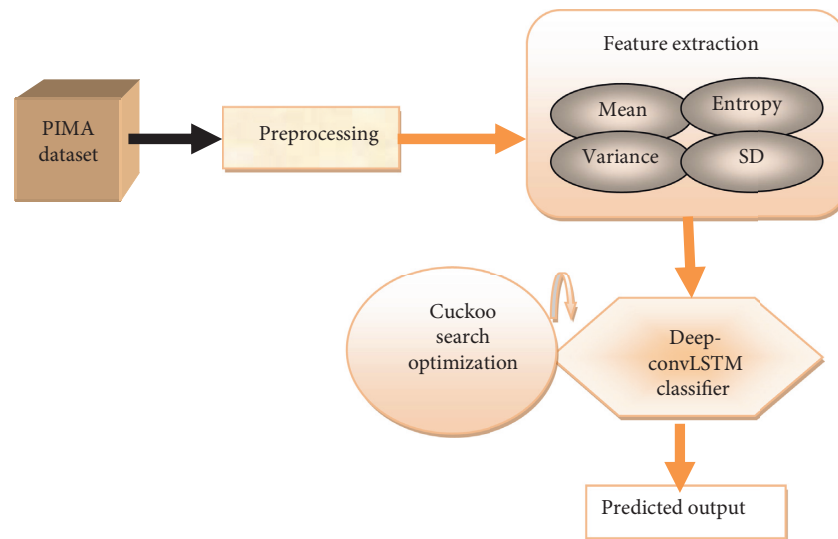


FIGURE 1: Disease prediction model using optimized convLSTM.

which obtained better classification accuracy. However, the updation of the fuzzy rules, and hence, the accuracy of the prediction is compromised under certain criteria. The estimation of the properties using the data was modeled by [20] using the optimization-based deep learning technique and obtained a more accurate estimation, however, the feature extraction is not considered, which may enhance the computational complexity of the system. The disease classification using the digital computer was developed by [21] using the segmentation and feature extraction with the support vector machine (SVM) but failed to enhance the accuracy by tuning the kernel function of the classifier. The ensemble-based classification of the disease was designed by [22, 23]. They obtained better classification accuracy using the most significant information selection but failed to consider the optimization that enhances the accuracy further. The healthcare monitoring using the IoMT was developed by [24] for the disease diagnosis. They obtained enhanced classification accuracy but failed to consider some of the significant features that may enhance the classification accuracy. IoT (Internet of things) is a combination of various communication devices and smart electronics that sense and communicate with each other. In recent years, IoT devices brought a revolution in the field of biomedical applications by looking at several challenges and complications faced in the past [33]. These IoT devices can generate a significant amount of biomedical data and also play a vital role in the development of existing automatic medical-data collection systems. When IoT devices are integrated with advanced ML (machine learning) algorithms, big data is essential for improvising these health systems in diagnosis, decision making, and treatment. IOT in biomedical applications has developed research areas in applications of IOE (Internet of everything), such as symptomatic treatments, observation of patients, and monitoring [25]. Additionally, the innovation of miniaturized healthcare sensors in monitoring the vital signs of the patient provided more security to the human healthcare system, and it offers more potential for early

diagnosis and treatment. The security issues in IoT devices are bound to increase gradually because of the expeditious development and deployment of IoT systems. The importance of security further increases in H-IoT because security breach in it can lead to the loss of lives. Various pieces of research are employed for the accurate disease prediction using IoT devices, however, in a broader perspective, it is quite difficult to accurately predict the disease most of the times. Basically, the prediction is performed using the empirical and dynamic methods. The empirical approach utilizes the previous information or historical prediction, and this approach is most commonly used in the regression and artificial neural networks. The dynamic approach is utilized by the physical and statistical methods. The advancement of technology in recent years promotes disease prediction using the techniques of regression, support vector machine (SVM), and K-nearest neighbor (KNN). Deep learning models are useful in examining large datasets, and they provide factual information. It is useful in computational applications [34]. Numerous methods are used in disease prediction and are categorized into three groups, namely statistical, dynamic, and satellite-based methods [35, 36], and the statistical methods are frequently used because of their inexpensive and time-consuming nature [37]. In this context, the review of the existing prediction models with the challenges of the research is presented, which motivated the researchers in designing a prediction model based on ANN and optimization. Kaur et al. [38] presented disparate ML techniques that were aimed at real-time along with remote HM on IoT infrastructure and associated with cloud computing. In their proposed approach, it uses numerous input attributes associated with that disease. The prediction systems were employed in evaluating certain diseases, for instance, HD, breast cancer, liver disorders, diabetes, thyroid, dermatology, spect_heart, along with surgical data. Mohan et al. [39] proposed a method that found significant good features by applying ML techniques, resulting in enhancing the accuracy on the forecast of cardiovascular disease.

Shailendra Tiwari et al. proposed a hybrid-cascaded framework for image reconstruction [40–43].

2.2. Challenges. The vulnerabilities gone through by the researchers are as follows:

- (i) Even though rapid miners are widely known for their data flow structure, the automatic optimization of larger medical data is vulnerable [1].
- (ii) It is challenging to combine the larger medical information in the form of websites or applications with higher detection accuracy [1].
- (iii) One of the main challenges is to get higher disease prediction performance for larger datasets on which most of the classifier depends [2].

The challenges faced by the existing disease prediction techniques are the failure to consider the preprocessing and postprocessing techniques that enhance the quality of prediction. Besides, the insignificant feature selection degrades the classification accuracy, and the failure to consider the same may enhance the computational complexity. Also, the failure to consider the optimization in the deep learning techniques reduces the classification accuracy. The above-mentioned challenges were solved using the proposed cuckoo search optimization based deep convolutional LSTM as it utilizes the pre-processing, most significant feature selection and optimization based deep learning mechanism to enhance the classification accuracy.

3. Proposed Disease Prediction Model Using Cuckoo Search-Based Deep Convolutional LSTM Classifier

Disease detection is essential to prevent serious health problems. Nowadays, IoT merged with deep learning methods is widely employed in various sectors. Hence, the healthcare sector is trying to implement IoT-based devices with the patients to effectively track the activities and disease-related capture for suggesting an effective diagnosis. Hence, in this research, a cuckoo search-based deep LSTM classifier is proposed for disease prediction for which the PIMA dataset [13] is used, which is preprocessed to remove the unwanted data, such as a negative value or infinity values. The processed data is then fed to the feature extractor, which draws only the appropriate features for evaluating the data. The extracted features are then sent to the classifier, where the cuckoo search algorithm is integrated to tune the biases and weights in the classifier. Figure 2 presented the disease prediction model using optimized convLSTM.

3.1. Read the Input Data. Initially, the data is obtained from the PIMA dataset [13], which is from the National Institute of Diabetes and Digestive and Kidney Diseases. The data is acquired from female patients above the age of 21 years and from an Indian background. The dataset is a combination of

independent variables, such as insulin level, age, BMI index, and one dependent variable.

3.2. Preprocessing the Input Data. Preprocessing the data is one of the main processes to improve the performance of the method. It also transforms the raw data into processable data. Here, to execute the healthcare PIMA dataset, the missing value imputation method is employed, which removes the infinity values for efficient processing.

3.3. Feature Extraction. Feature extraction is the process of extracting the significant and essential data for prediction, which assures the effective presentation of the raw input data. In this research, feature extraction, including the statistical features, such as variance, mean, standard deviation, and entropy, are acquired.

3.3.1. Mean. Mean is a statistical feature, which is the ratio of the sum of the data to the total number of the data present in the database. It is mathematically represented as follows:

$$\text{mean} = \frac{1}{a} \sum_{t=1}^a R_t, \quad (1)$$

where a indicates the total instances and R_t indicates the t^{th} data.

3.3.2. Variance. The variance is the mean squared difference between each data and the mean, which is given by the following:

$$\text{variance} = \frac{1}{a-1} \sum_{t=1}^a (R_t - \text{mean})^2. \quad (2)$$

At different instances of time, the variance concentrates on minute changes, which enhances the prediction accuracy.

3.3.3. Standard Deviation (SD). Standard deviation shows the amount of variation or dispersion that exists at each instance of the data concerning mean. It is measured as the root of variance, which is given as follows:

$$\text{standard deviation} = \sqrt{\frac{1}{a-1} \sum_{t=1}^a (R_t - \text{mean})^2}. \quad (3)$$

3.3.4. Entropy. Entropy is the probability of the total number of ways the data can be arranged, which is formulated as follows:

$$\text{entropy} = - \sum_{x=1}^A b_x \log_2 b_x, \quad (4)$$

where b_x denotes the probability of the data, and x indicates the possible sample values.

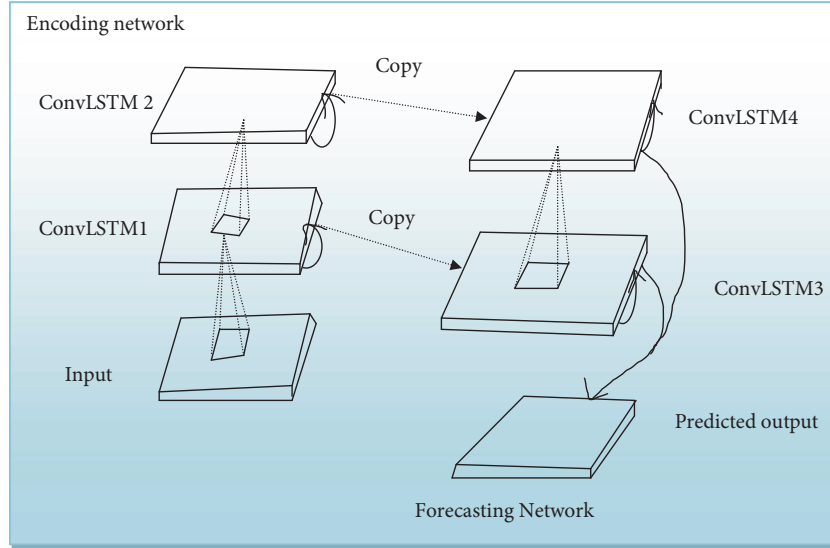


FIGURE 2: Architecture of deep-ConvLSTM.

3.4. Disease Prediction Using the Proposed Cuckoo-Based Deep-ConvLSTM Classifier. Disease prediction at the early stage is essential to improve the health of the patients. Hence, in this research, a deep learning model is implemented in the IoT systems for the efficient prediction of diseases. In the deep convolutional LSTM classifier, the convolution operation is merged with the long-short term memory. Here, the multiplication operation is replaced with a convolution operation, which records the essential spatial features for effective disease prediction. The cuckoo search algorithm is employed for tuning the weights and bias of the deep-convLSTM classifier. In this optimization, this bird has the best communication nature and chooses only the best eggs, and the selection characteristics of the cuckoo are integrated with the classifier in the internal modal parameter tuning that improves the prediction performance.

3.4.1. Architecture of Deep-ConvLSTM Classifier. Figure 1 shows the architectural explanation for the deep Conv LSTM classifier. The Conv LSTM is arranged as 3D tensors with $F_1, \dots, F_\beta$ as inputs, $G_1, \dots, G_\beta$ as cell outputs, $K_1, \dots, K_\beta$ as the hidden states, and $H_\beta, I_\beta, J_\beta$ as the gates of the Conv LSTM. Conv LSTM uses a large transitional kernel for efficient feature patterns. For efficient prediction, they are assembled as encoding layers and forecasting layers. The cell output and the initial states of the encoding layer are copied to the forecasting network. The dimensions of the predicted output are the same as the input. Hence, they are combined and sent to the $[1 \times 1]$ convolutional layer for the final prediction of output. The output of the input gate is expressed as follows:

$$H_\beta = \alpha(\eta_H^F * F_\beta + \eta_H^K * K_{\beta-1} + \eta_H^G \circ G_{\beta-1} + \gamma^H). \quad (5)$$

Here, F_β denotes the input vector, η_H^F denotes the weight between the input layer and input gate, α is the gate activation function, η_H^K denotes the weight between the input

layer and the memory output, η_H^G denotes the input layer and the cell output, and $K_{\beta-1}, G_{\beta-1}$ are the preceding outputs of the cell and memory unit, respectively. γ^H denotes the bias of the input layer, $*$ is the convolutional operator, and $\circ$ is the element-wise multiplication. The forget gate output is expressed as follows:

$$I_\beta = \alpha(\eta_I^F * F_\beta + \eta_I^K * K_{\beta-1} + \eta_I^G \circ G_{\beta-1} + \gamma^I). \quad (6)$$

Here, η_I^F denotes the weight in between the input layer and the forget gate, η_I^K denotes the weights for the connections in between the output gate and the memory units of the previous layers, η_I^G denotes the weight in between the output gates and the cell, and γ^I is the bias regarding the forget gate. The result of the output gate is expressed as follows:

$$J_\beta = \alpha(\eta_J^F * F_\beta + \eta_J^K * K_{\beta-1} + \eta_J^G \circ G_{\beta-1} + \gamma^J). \quad (7)$$

Here, η_J^F indicates the weight in between the output gate and the input layer, η_J^K indicates the weight in between the output gate and the memory unit, η_J^G indicates the weight in the middle of the output gate and the cell, and γ^J is the output gate. The output of the temporary cell state is formulated as follows:

$$\tilde{G}_\beta = \tanh(\eta_L^F * F_\beta + \eta_L^K * K_{\beta-1} + \gamma^L). \quad (8)$$

Here, η_L^F denotes the weight in the middle of the cell and the input layer, η_L^K denotes the weight in between the cell and the memory unit, and γ^L is the bias of the cell. The output of the cell is given as follows:

$$\begin{aligned} G_\beta &= I_\beta \circ G_{\beta-1} + H_\beta \circ \tilde{G}_\beta \\ &= I_\beta \circ G_{\beta-1} + H_\beta \circ \tanh(\eta_L^F * F_\beta + \eta_L^K * K_{\beta-1} + \gamma^L). \end{aligned} \quad (9)$$

The output from the memory unit is expressed as follows:

$$K_\beta = J_\beta \circ \tan h(G_\beta). \quad (10)$$

Here, K_β is the output of the memory block, and J_β denotes the output gate. The output of the output layer is expressed as follows:

$$N_\beta = \lambda(\eta_N^K \cdot K_\beta + \gamma^N), \quad (11)$$

where η_N^K denotes the weight between the output vector and the memory unit, and γ^N is the bias of the output layer. Hence, the bias and the weights are represented as follows:

$$\begin{aligned} \gamma &\in \{\gamma^L, \gamma^I, \gamma^J, \gamma^H\}, \\ \eta &\in \{\eta_N^K, \eta_L^F, \eta_L^K, \eta_J^F, \eta_J^K, \eta_I^G, \eta_I^F, \eta_I^K, \eta_I^G, \eta_H^F, \eta_H^K, \eta_H^G\}. \end{aligned} \quad (12)$$

3.4.2. Mathematical Model for Cuckoo Search Optimization. The weights η and bias γ for the deep Conv LSTM classifier are generated using the cuckoo search algorithm, where the cuckoos are known for their wonderful voices and their breeding process. They lay their eggs in the nest of other birds, which resemble their eggs. Hence, the host bird considers the cuckoo eggs to be their eggs and feeds them. Thus, the cuckoos protect their eggs from dying, and when eggs are hatched, the cuckoo chick throws out the host eggs from the nest. Hence, they receive more food from the host nest, however, if the host bird discovers these eggs, they throw these eggs or leave their nest and build a new one [12]. There are three basic rules in cuckoo search optimization, which are as follows:

- (1) Cuckoos choose a host nest for preserving their eggs.
- (2) Nests with high quality have good eggs that are taken to the future generation.
- (3) The total number of the host nests will be decided along with the probability that the host can find the cuckoo eggs.

Population initialization: cuckoos initialize the chosen nests for preserving their hatched eggs. The objective function is indicated as, $d(g), g = (g_1, \dots, g_i)$.

The equation for initializing the host nest is expressed as, $g_j (j = 1, 2, \dots, y)$, where y denotes the nest.

Fitness evaluation: fitness evaluation is carried out on the basis of accuracy. If the obtained fitness value is greater than the previous iteration, then the obtained value is chosen as the best solution, which is given as the condition $Q_j > Q_v$, where Q denotes the fitness, and v is the randomly chosen nest in y .

Position updating: the generation of new solutions, $g_j^{(k+1)}$, for cuckoo j , a levy flight can be carried out.

$$g_j^{(k+1)} = g_j^{(k)} + \mu \oplus \text{Levy}(\theta), \quad (13)$$

where $\mu > 0$ denotes the step size of the problem, and $\oplus$ denotes the entry-wise multiplication.

Levy distribution for a large number of steps is expressed as follows:

$$\text{Levy} \sim h = k^{-\theta}, \quad (1 < \theta \leq 3). \quad (14)$$

Termination: cuckoos find the eggs of other birds in the nest, with the probability given as $P_c \in [0, 1]$. The cuckoos discard them from the nest and build a new nest by updating their position through levy flights, thereby repeating the iteration until it reaches the best fitness solution to generate a global optimal solution. Algorithm 1 presented the pseudocode for the proposed cuckoo-based deep convLSTM.

Algorithm 1: the pseudocode for the proposed cuckoo-based deep convLSTM.

S.NO Pseudocode for the proposed cuckoo-based deep convLSTM

- (1) Input: g_j
- (2) Output: $g_{j,\text{best}}$
- (3) Initialize the population
- (4) Fitness evaluation termination
- (5) If $Q_j > Q_v$
- (6) Replace the new solution
- (7) end
- (8) While $k < \text{maximum generation}$
- (9) New solution generated using the equation (13) and (12)
- (10) Re-evaluate the fitness
- (11) New optimal global solution, g_{best}
- (12) End while
- (13) end

4. Results and Discussion

This section illustrates the experimental analysis and the comparative results acquired while using the proposed cuckoo-based deep convLSTM model.

4.1. Experimental Setup. The MATLAB tool is used for the implementation of the proposed method for predicting diseases. The MATLAB tool is established in the Windows 10 OS and 64 bit operating systems with 16 GB RAM, which provides an efficient and simple implementation of the proposed method.

4.2. Data Description. PIMA dataset is employed for the prediction of the diabetic disease. Originally, this dataset is obtained from the National Institute of Diabetes and Digestive and Kidney Diseases. The dataset includes female patients above 21 years with an Indian background. The dataset is the combination of independent variables, such as insulin level, age, BMI index, and one dependent variable. The sample record is depicted in Figure 3.

4.3. Evaluation Metrics. The disease prediction ability of the proposed cuckoo-based deep convLSTM is analyzed based

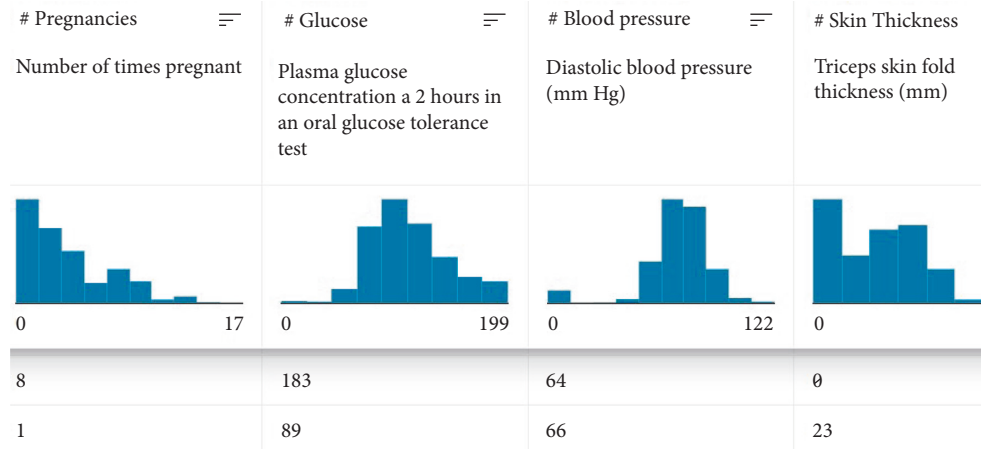


FIGURE 3: Sample record of database.

on accuracy, sensitivity, and specificity. They are enumerated as follows:

4.3.1. Accuracy. Accuracy is the nearness of the predicted output obtained by the proposed method to the standard value, which is mathematically expressed as follows:

$$\text{accuracy} = \frac{\text{measured positive} + \text{measured negative}}{\text{true positive} + \text{true negative}}. \quad (15)$$

4.3.2. Sensitivity. Sensitivity is the ability to identify patients with the diabetic disease correctly by the proposed method for efficient prediction, which is mathematically formulated as follows:

$$\text{sensitivity} = \left(\frac{\text{true positives}}{\text{total diseased}} \right). \quad (16)$$

4.3.3. Specificity. Specificity is the ability to identify patients with nondiabetic diseases correctly by the proposed method, which is mathematically formulated as follows:

$$\text{specificity} = \left(\frac{\text{true negative}}{\text{total real negative cases}} \right). \quad (17)$$

4.4. Comparative Analysis. In this section, the comparative analysis of the proposed cuckoo-based deep convLSTM is deployed for the efficient detection of diabetic disease. The methods considered for comparative analysis are ANN [14], CNN [15], deep convLSTM [16], MDCNN [23], and MSSO-ANFIS [24].

4.4.1. Comparative Analysis in terms of the Training Percentage. The proposed cuckoo-based deep Conv LSTM is compared with the existing conventional methods, such as ANN, CNN, and deep convLSTM. Figure 4(a) illustrates the analysis concerning the accuracy. The accuracy of the

conventional methods and the proposed methods in terms of the training percentage of 80% are 91.99%, 95.56%, 82.96%, 87.02%, 94.17%, and 96.95%, respectively, for the methods, such as MSSO-ANFIS, MDCNN, ANN, CNN, Deep-Conv LSTM, and proposed Cuckoo-based Deep-Conv LSTM. Figure 4(b) illustrates the analysis with respect to sensitivity. The sensitivity of the conventional methods and the proposed methods in terms of the training percentage of 70% are 88.56%, 92.08%, 80.61%, 83.73%, 90.78%, and 93.39%, respectively, for the methods, such as MSSO-ANFIS, MDCNN, ANN, CNN, Deep-Conv LSTM, and proposed Cuckoo-based Deep-Conv LSTM. Figure 4(c) illustrates the analysis with respect to the specificity. The specificity of the conventional methods and the proposed methods in terms of the training percentage of 40% are 89.05%, 90.70%, 83.34%, 86.80%, 90.09%, and 91.30%, respectively, for the methods, such as MSSO-ANFIS, MDCNN, ANN, CNN, Deep-Conv LSTM, and proposed Cuckoo-based Deep-Conv LSTM.

4.4.2. Comparative Analysis in terms of K-fold. The proposed cuckoo-based deep convLSTM is compared with the existing conventional methods, such as ANN, CNN, and deep convLSTM. Figure 5(a) illustrates the analysis with respect to the accuracy. The accuracy of the conventional methods and the proposed methods in terms of the k-fold = 5 are 79.80%, 80.39%, 75.62%, 78.74%, 79.92%, and 80.87%, respectively, for the methods, such as MSSO-ANFIS, MDCNN, ANN, CNN, Deep-Conv LSTM, and proposed Cuckoo-based Deep-Conv LSTM. Figure 5(b) illustrates the analysis with respect to the sensitivity. The sensitivity of the conventional methods and the proposed methods in terms of the k-fold of 7 are 77.83%, 78.44%, 73.49%, 76.04%, 77.25%, and 79.63%, respectively, for the methods, such as MSSO-ANFIS, MDCNN, ANN, CNN, Deep-Conv LSTM, and proposed Cuckoo-based Deep-Conv LSTM. Figure 5(c) illustrates the analysis with respect to specificity. The specificity of the conventional methods and the proposed methods in terms of the k-fold of 9 are 88.26%, 88.38%, 85.80%, 87.63%, 87.86%, and 88.90%, respectively,

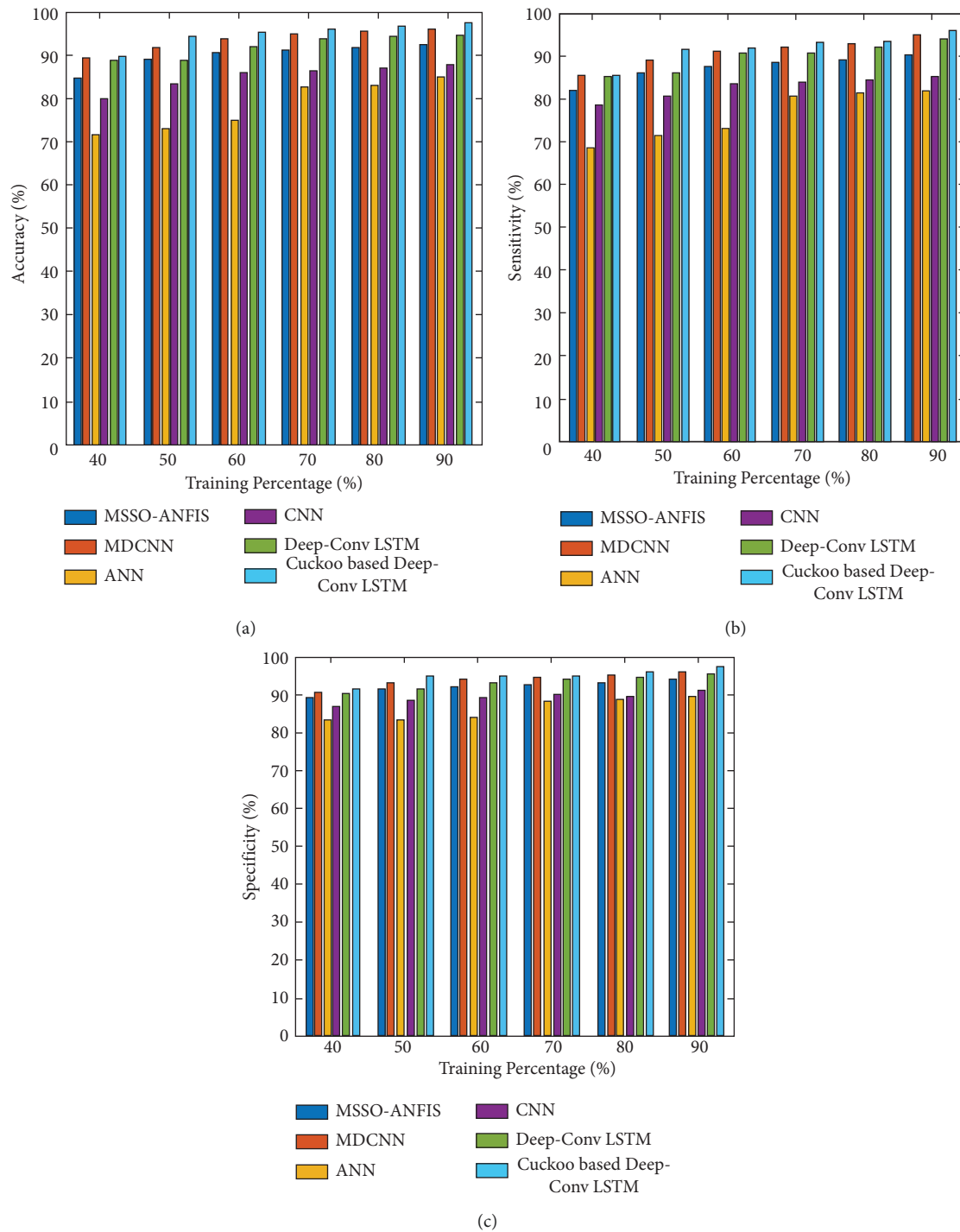


FIGURE 4: Analysis of the methods based on training percentage, (a) accuracy, (b) sensitivity, and (c) specificity.

for the methods, such as MSSO-ANFIS, MDCNN, ANN, CNN, Deep-Conv LSTM, and proposed Cuckoo-based Deep-Conv LSTM.

4.5. Comparative Discussion. Table 1 depicts the comparative Analysis of the proposed method, in which the best performance is obtained by the proposed method in terms of the performance metrics by varying the training and the

K-Fold values. The maximal accuracy obtained by the proposed method is 97.59%, which shows 5.01%, 1.46%, 12.74%, 10.03%, and 2.92% superior performance as compared to the existing MSSO-ANFIS, MDCNN, ANN, CNN, and Deep-Conv LSTM methods, respectively. The maximal sensitivity obtained by the proposed method is 95.87%, which shows 5.63%, 0.99%, 14.55%, 11.26%, and 1.98% superior performance as compared to the existing MSSO-ANFIS, MDCNN, ANN, CNN, and Deep-Conv LSTM

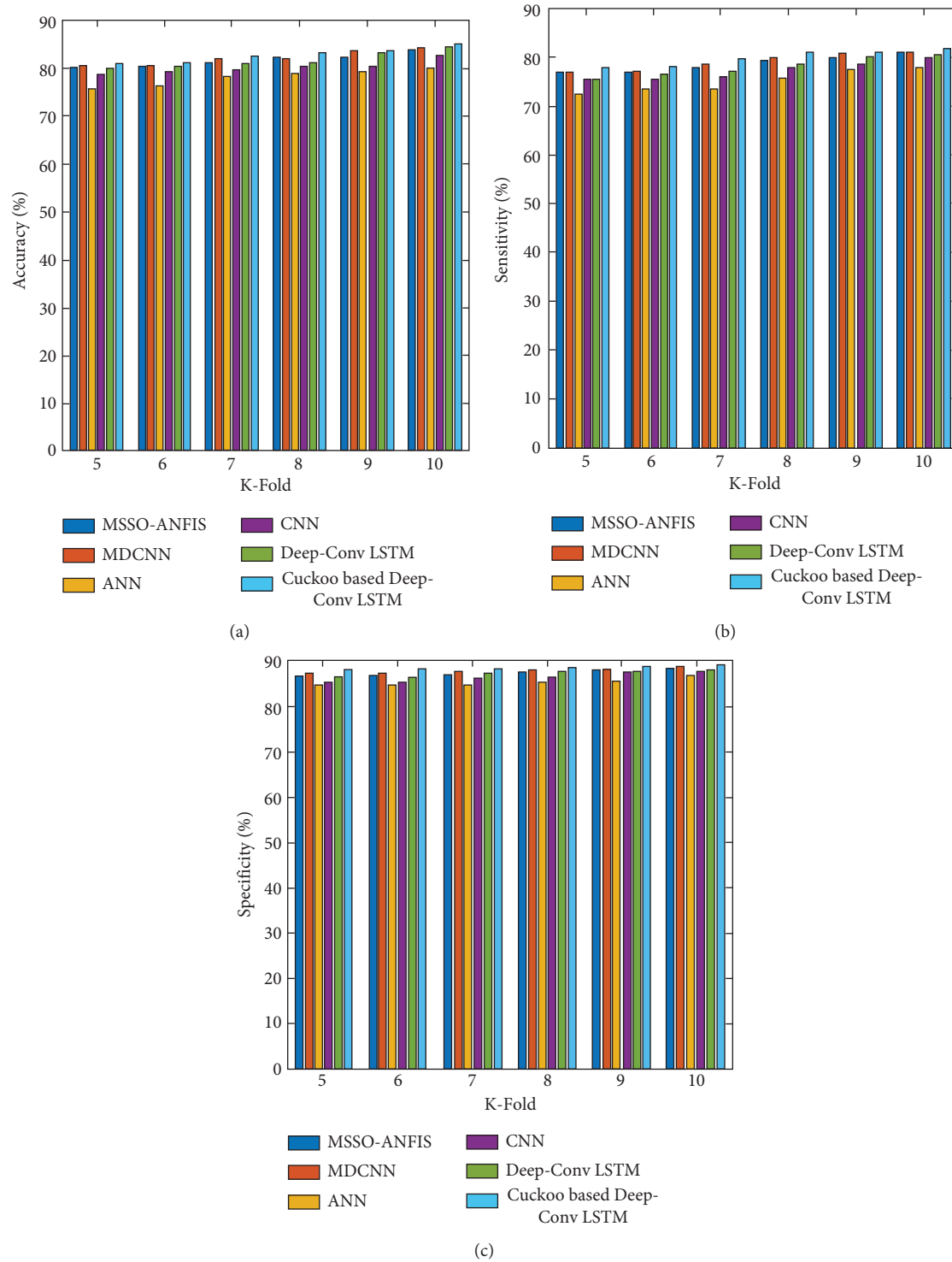


FIGURE 5: Analysis of the methods based on k-fold. (a) Accuracy, (b) sensitivity, and (c) specificity.

TABLE 1: Comparative analysis.

Method/Metrics		MSSO-ANFIS	MDCNN	ANN	CNN	Deep-conv LSTM	Proposed cuckoo-based deep-conv LSTM
Training percentage	Accuracy (%)	92.70	96.17	85.15	87.80	94.74	97.59
	Sensitivity (%)	90.47	94.92	81.92	85.08	93.97	95.87
	Specificity (%)	94.07	96.13	89.14	91.04	95.17	97.09
K-fold	Accuracy (%)	83.95	84.67	80.03	82.72	84.17	85.18
	Sensitivity (%)	80.86	81.12	77.78	79.97	80.50	81.74
	Specificity (%)	88.53	88.63	86.80	87.85	88.04	89.22

methods, respectively. The maximal specificity obtained by the proposed method is 97.09%, which presents 3.11%, 0.98%, 8.19%, 6.23%, and 1.97% superior performance as compared to the existing MSSO-ANFIS, MDCNN, ANN, CNN, and Deep-Conv LSTM methods, respectively.

The proposed method obtained elevated performance compared to the existing techniques in terms of performance metrics. It is because the proposed method utilizes the preprocessing of the input data, which removed the noises and artifacts. Then, the most significant feature extraction reduces the computational complexity by eliminating the less informative features. Finally, the diabetic disease prediction using the proposed cuckoo search optimization-based Deep-Conv LSTM enhances the prediction accuracy through the optimal tuning of weights and biases.

5. Conclusion

The proposed cuckoo-based deep convLSTM model is deployed for the prediction of diseases with higher efficiency. The data from the PIMA dataset is used and for effective prediction missing value imputation method is employed for processing the data. The cuckoo search optimization, which is a nature-inspired algorithm, is deployed with a deep convLSTM classifier as it effectively predicts the disease by transferring the information, thus reducing time consumption. A deep convLSTM classifier enhances the disease prediction ability through a convolution operation based on the weights and features, which generates highly refined information regarding the input data. The proposed method is compared with the conventional methods and revealed that the proposed method achieved 97.591%, 95.874%, and 97.094% of accuracy, sensitivity, and specificity, respectively. In the future, more datasets and highly advanced algorithms will be designed to further boost the classifier performance.

Data Availability

Data will be available on request. For the data related queries, kindly contact Ashwani Kumar at ashwani.kumarcse@gmail.com.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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